Pre-work · Appendicular Skeleton

Appendicular Skeleton, the upper and lower limbs. Read both Part A chapters first. Learn the markings vocabulary before the markings, because it repeats on every bone.

Module 2 · 3 of 4 | Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

1. **Read**
Read the Part A chapter first. This packet does not repeat it. It puts it to work.
2. **Organize**
Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
3. **Draw**
Panel 2. Brain dump from memory, then draw, then correct in a second colour.
4. **Apply**
Panel 3. Notes closed for the first pass. Open them only to check.
5. **Retrieve**
Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the two limb chapters into mini visuals. Eight chunks, and the two comparison splits are worth the most.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A patient falls on an outstretched hand and has tenderness in the anatomical snuffbox. Name the likely fracture and say why it matters.

Answer. A scaphoid fracture.

Reasoning. Start from the location, then ask what lies under it. The snuffbox is the hollow at the base of the thumb, and the scaphoid sits in its floor, so tenderness there points at that bone. It matters because the scaphoid has a poor blood supply, so a fracture missed on a first X-ray can fail to heal. That is a structural fact producing a clinical rule: suspect it, splint it, image it again.

Set A · Name the marking

1. The only bony attachment of the upper limb to the axial skeleton is the ____________________.

takes longer than three minutes you are copying instead of organizing.

1 **GRID**

Bone markings vocabulary

Marking down the side, type and example across the top. Nine rows. Sort each as a projection, a depression, or an opening. Do this one first.

2 **CHAIN**

Upper limb, proximal to distal

Girdle, arm, forearm, hand, with the bones under each link. Then run a second chain underneath for the lower limb and line the two up so the parallels show.

3 **HUB**

The scapula

Scapula at the hub. Spoke to the three borders, the spine, acromion, coracoid process, glenoid cavity, and the three fossae.

4 **SPLIT**

Radius against ulna

One split. Side of the forearm, proximal markings, distal markings on each branch. This is the pair most people lose.

5 **HUB**

The humerus

Humerus down the middle. Proximal markings on one side, distal on the other. Mark which distal surface meets which forearm bone.

6 **HUB**

The coxal bone

Acetabulum at the hub, because all three bones meet there. Spoke out to ilium, ischium, and pubis, then to the markings on each.

DRAWING 1

The humerus, both views

BRAIN DUMP FIRST

Two minutes. Every marking on the humerus, proximal and distal.

Draw the humerus twice, anterior and posterior. On the anterior view mark head, anatomical and surgical necks, greater and lesser tubercles, intertubercular sulcus, deltoid tuberosity, capitulum, trochlea, and both epicondyles. On the posterior view add the olecranon fossa and the radial groove, and mark where the radial nerve runs.

In your notebook, head the page **M2–1 The humerus, both views** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can say which forearm bone meets the capitulum and which meets the trochlea without pausing.

2. The socket that receives the head of the humerus is the ______________________.
3. The point of the elbow is the ______________________ of the ______________________.
4. The deep cup where the three hip bones meet is the ______________________.
5. The tarsal that receives the body weight from the tibia is the ______________________.

Set B · Predict it

1. A fracture at the surgical neck of the humerus. Name the nerve at risk and say why it is checked.

   ____

   ____

2. A patient cannot feel the little finger after knocking the medial elbow. Name the nerve and where it runs.

   ____

   ____

3. An older adult falls and cannot bear weight on one leg, with the limb shortened and rotated. Name the likely fracture site and the structure whose blood supply is threatened.

   ____

   ____

4. Surgeons harvest bone from the fibula rather than the tibia. Explain why, using weight bearing.

   ____

   ____

Set C · Tell it apart

1. Give one feature that tells the radius from the ulna at the elbow, and one at the wrist.

   ____

   ____

2. Name the carpals in both rows in order, using the mnemonic, and say which is fractured most often.

   ____

   ____

7 SPLIT

Tibia against fibula

One split. Side, whether it bears weight, and the malleolus each one forms. Three branches, and the weight-bearing branch explains the rest.

8 GRID

Carpals and tarsals

Two grids side by side. Carpals in two rows of four with the mnemonic, tarsals in their arrangement with the three C's around the N and T.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 2

The forearm in supination and pronation

BRAIN DUMP FIRST

One minute. Radius and ulna, which is which, and what moves.

Draw the two bones side by side in supination, palm forward, and label radius lateral and ulna medial. Then draw them again in pronation with the radius crossed over the ulna. Mark the proximal and distal radioulnar joints and the interosseous membrane in both.

In your notebook, head the page **M2–2 The forearm in supination and pronation** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can explain which bone moves and which stays put, and name the joints that allow it.

3. Compare the pectoral and pelvic girdles by stability and range of motion, and say what causes the difference.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

DRAWING 3

The coxal bone and the knee

BRAIN DUMP FIRST

One minute. The three fused bones, then the ligaments of the knee.

Draw the coxal bone from the side, shading ilium, ischium, and pubis differently, and mark the acetabulum where all three meet plus the obturator foramen. Then draw the knee from the front with the patella removed: femoral condyles, tibial plateau, both menisci, both cruciate ligaments crossing, and both collateral ligaments.

In your notebook, head the page **M2–3 The coxal bone and the knee** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can name the three structures of the unhappy triad on your own knee drawing and say what force causes it.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 Bone markings vocabulary

GRID

2 Upper limb, proximal to distal

CHAIN

3 The scapula

HUB

4 Radius against ulna

SPLIT

5 The humerus
HUB

6 The coxal bone
HUB

7 Tibia against fibula
SPLIT

8 Carpals and tarsals
GRID